

FAULT CODE 4615 - Fuel Pump Delivery Pressure - Data Valid But Above Normal Operating Range - Most Severe Level

Warnings and Cautions

CAUTION

To reduce the possibility of damaging a new engine control module (ECM), all other active fault codes must be investigated before replacing the ECM.

CAUTION

To reduce the possibility of pin and harness damage, use the following test leads when taking a measurement: Part Number 3822758 - male Deutsch™/AMP™/Metri-Pack™ test lead and Part Number 3823993 - male Deutsch™ test lead.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the low-pressure fuel system.	
	STEP 1A. Check the fuel supply pressure sensor.	INSITE™ electronic service tool and pressure gauge readings within 14 kPa [2 psi] of each other?
	STEP 1B. Check the fuel supply lines.	Any fuel lines damaged, kinked, or restricted?
	STEP 1C. Check the Stage 2 fuel filter restriction.	Stage 2 fuel filter restriction within specification?
	STEP 1C-1. Check the fuel filter head.	Stage 2 fuel filter restriction within specification?
	STEP 1D. Check the fuel supply pressure at the fuel inlet manifold.	Fuel inlet pressure always less than 34.5 kPaG [5 psig]?

STEPS		SPECIFICATIONS
STEP 2.	Disable and clear the fault code.	
	STEP 2A. Disable the fault code.	Fault Code 4615 inactive?
	STEP 2B. Clear the inactive fault codes.	All fault codes cleared?

STEP 1. Check the fuel supply pressure fuel system.

STEP 1A. Check the fuel supply pressure sensor.

- Conditions :
 - Connect the Compuchek™ tool.
 - Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
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Check the fuel supply pressure sensor. - Start and operate the engine at high idle. - Record the fuel supply sensor reading in INSITE™ electronic service tool. - Record the fuel supply pressure reading on a pressure gauge installed in place of the fuel supply pressure sensor. - Use the following procedure for Stage 2 filters in the Measure section of the K38, K50, QSK38 and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 006-024 in the Associated Procedures Table. - Use the following procedure for Stage 2 filters in the Measure section in the QSK45 and QSK60 Service Manual, Bulletin 4021530, . Refer to Procedure 006-024 in the Associated Procedures Table.	INSITE™ electronic service tool and pressure gauge readings within 14 kPa [2psia] of each other? YES	1B
	INSITE™ electronic service tool and pressure gauge readings within 14 kPa [2psia] of each other? NO Repair : Replace the fuel supply pressure sensor. Refer to Procedure 019-398 in the Associated Procedures Table.	2A

STEP 1B. Check the fuel supply lines.

Conditions :

- Turn keyswitch OFF.

Action	Specification/Repair	Next Step
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Examine the fuel supply lines for damages, kinks, or restrictions. Refer to Procedure 006-024 in the Associated Procedures Table.	Any fuel lines damaged, kinked, or restricted? YES Repair : Replace the damaged fuel line(s).	2A
	Any fuel lines damaged, kinked, or restricted? NO	1C

STEP 1C. Check the Stage 2 fuel filter restriction.

Conditions : - Turn keyswitch OFF. - Connect INSITE™ electronic service tool.		
Action	Specification/Repair	Next Step

Measure the Stage 2 filter restriction. • Use the following procedure in the K38, K50, QSK38 and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 006-020 in the Associated Procedures Table. • Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 006-020 in the Associated Procedures Table.	Stage 2 fuel filter restriction within specification? YES	1D
	Stage 2 fuel filter restriction within specification? NO Repair : Replace the Stage 2 fuel filters. • Use the following procedure in the K38,K50, QSK38 and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 006-076 in the Associated Procedures Table. • Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 006-076 in the Associated Procedures Table.	1C-1

STEP 1C-1. Check the fuel filter head.

Conditions :

- Turn keyswitch OFF.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step

Measure the Stage 2 fuel filter restriction. • Use the following procedure in the K38, K50, QSK38 and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 006-020 in the Associated Procedures Table. • Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 006-020 in Associated Procedures Table.	Stage 2 fuel filter restriction within specification? YES	2A
	Stage 2 fuel filter restriction within specification? NO Repair : Replace the Stage 2 fuel filter head. • Use the following procedure in the K38, K50, QSK38 and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 006-020 in the Associated Procedures Table. • Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 006-020 in Associated Procedures Table.	2A

STEP 1D. Check the fuel supply pressure at the fuel inlet manifold.

Conditions :

- Turn keyswitch OFF.

Action	Specification/Repair	Next Step
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Check the fuel supply pressure at the fuel manifold. • Close the fuel supply valve. • Remove the M14 STOR plug from the fuel inlet manifold. • Install a Compuchek™ fitting, Part Number 3824844, or equivalent, in the fuel inlet manifold. • Connect the vacuum gauge and adapter, Part Number 3164491, or equivalent, and digital multimeter, Part Number 3164488 or 3164489, or equivalent, to the Compuchek™ fitting.	Fuel inlet pressure always less than 34.5 kPaG [5 psig]? YES Repair : Replace the fuel pump. • Use the following procedure the K38, K50, QSK38 and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 005-016 in the Associated Procedures Table. • Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 005-016 in the Associated Procedures Table.	2A
• Open the fuel supply valve. • Start and operate the engine at low idle and high idle. • Record the fuel inlet pressure readings.	Fuel inlet pressure always less than 34.5 kPaG [5 psig]? NO Repair : Refer to the equipment manufacturer service information to reduce fuel inlet pressure.	2A

STEP 2. Disable and clear the fault code.

STEP 2A. Disable the fault code.

Conditions :

- Connect all components.
- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.
- Operate the engine.

Action	Specification/Repair	Next Step
Disable the fault code. • Use INSITE™ electronic service tool to verify the fault code is inactive.	Fault Code 4615 inactive? YES	2B
	Fault Code 4615 inactive? NO Repair : Return to the troubleshooting steps or contact a Cummins® Authorized Repair Location if all steps have been completed and checked again.	1A

STEP 2B. Clear the inactive fault codes.**Conditions :**

- Connect all components.
- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Clear the inactive fault codes. • Use INSITE™ electronic service tool to clear the inactive fault codes.	All fault codes cleared? YES	Repair complete
	All fault codes cleared? NO Repair : Troubleshoot any remaining fault codes.	Go to the appropriate troubleshooting steps.

Associated Procedures

Associated Procedures			
Procedure Title	Procedure Number	Service Model Name	Bulletin Number
Fuel Pump	Refer to Procedure 005-016 (/qs3/pubsys2/xml/en/procedures/28/28-005-016-tr.html)	K38, K50, QSK38 and QSK50	4021528
Fuel Pump	Refer to Procedure 005-016 (/qs3/pubsys2/xml/en/procedures/56/56-005-016-tr.html)	QSK38 and QSK50	4021530
Fuel Filter Head	Refer to Procedure 006-017 (/qs3/pubsys2/xml/en/procedures/28/28-006-017-tr.html)	K38, K50, QSK38 and QSK50	4021528
Fuel Filter Head	Refer to Procedure 006-017 (/qs3/pubsys2/xml/en/procedures/56/56-006-017-tr.html)	QSK45 and QSK60	4021530
Fuel Inlet Restriction	Refer to Procedure 006-020 (/qs3/pubsys2/xml/en/procedures/28/28-006-020-tr.html)	K38, K50, QSK38 and QSK50	4021528
Fuel Inlet Restriction	Refer to Procedure 006-020 (/qs3/pubsys2/xml/en/procedures/56/56-006-020-tr.html)	QSK45 and QSK60	4021530
Fuel Supply Lines	Refer to Procedure 006-024 (/qs3/pubsys2/xml/en/procedures/28/28-006-024-tr.html)	K38, K50, QSK38, and QSK50	4021528

Associated Procedures			
Procedure Title	Procedure Number	Service Model Name	Bulletin Number
Fuel Supply Lines	Refer to Procedure 006-024 (/qs3/pubsys2/xml/en/procedures/56/56-006-024-tr.html)	QSK45 and QSK60	4021530
Fuel Filter (Stage 2)	Refer to Procedure 006-076 (/qs3/pubsys2/xml/en/procedures/28/28-006-076-tr.html)	K38, K50, QSK38, and QSK50	4021528
Fuel Filter (Stage 2)	Refer to Procedure 006-076 (/qs3/pubsys2/xml/en/procedures/56/56-006-076-tr.html)	QSK45 and QSK60	4021530
Fuel Supply Pump Pressure Sensor	Refer to Procedure 019-398 (/qs3/pubsys2/xml/en/procedures/122/122-019-398.html)	QSK19	4022094
Fuel Supply Pump Pressure Sensor	Refer to Procedure 019-398 (/qs3/pubsys2/xml/en/procedures/122/122-019-398.html)	QSK38, QSK50, and QSK60	4022102

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